eTable 1 Full search strategies for meta-analysis of studies reporting sex specific outcomes of patients with STEMI.

Database	Search strategy (publications accessible January 1, 2010 to August 1, 2020)
PubMed	("gender"[Title/Abstract] OR "female"[Title/Abstract] OR "male"[Title/Abstract] OR "gender differences"[Title/Abstract] OR "sex differences"[Title/Abstract] OR "sex characteristics"[MeSH Terms]) AND ("death"[MeSH Terms] OR "mortality"[MeSH Terms] OR "hospital mortality"[MeSH Terms] OR "cardiac death"[Title/Abstract] OR "sudden cardiac death"[MeSH Terms] OR "all-cause mortality"[Title/Abstract] OR "long term mortality"[Title/Abstract] OR "one year mortality"[Title/Abstract] OR "cardiovascular mortality"[Title/Abstract] OR "short term mortality"[Title/Abstract]) AND ("myocardial infarction"[MeSH Terms] OR "acute myocardial infarction"[Title/Abstract] OR "ST Elevation Myocardial Infarction"[MeSH Terms] OR "myocardial necrosis"[Title/Abstract] OR "primary percutaneous coronary intervention"[Title/Abstract] OR "primary PCI"[Title/Abstract] OR "primary angioplasty"[Title/Abstract])
EMBASE	(gender.mp OR female.mp OR male.mp OR gender differences.mp OR sex differences.mp OR sex characteristics.mp) AND (death.mp OR mortality.mp OR hospital mortality.mp OR cardiac death.mp OR sudden cardiac death.mp OR all-cause mortality.mp OR long term mortality OR one year mortality.mp OR cardiovascular mortality.mp OR short term mortality) AND (myocardial infarction.mp OR acute myocardial infarction.mp OR ST Elevation Myocardial Infarction.mp OR myocardial necrosis.mp OR primary percutaneous coronary intervention.mp OR primary PCI.mp OR primary angioplasty.mp)
Cochrane Library	[Title and abstract search] (gender OR female OR male OR gender differences OR sex differences OR sex characteristics) AND (death OR mortality OR hospital mortality OR cardiac death OR sudden cardiac death OR all-cause mortality OR long term mortality OR one year mortality OR cardiovascular mortality OR short term mortality) AND (myocardial infarction OR acute myocardial infarction OR ST Elevation Myocardial Infarction OR myocardial necrosis OR primary percutaneous coronary intervention OR primary PCI OR primary angioplasty)

eTable 2 Variables adjusted in the adjusted analyses from the included studies.

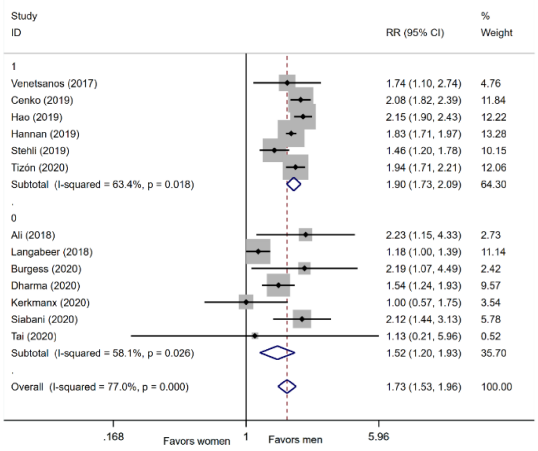
First Author	Year	Adjusted Variables
Venetsanos	2017	age, weight, prior MI, prior PCI, patient's history of diabetes, hypertension, non-hemorrhagic stroke, gastrointestinal bleeding, time from symptom onset to pre-PCI ECG, admission Killip class, baseline hemoglobin, eGFR, access site, use of Glycoprotein IIIb/IIa inhibitor, bivalirudin and unfractionated heparin, location of MI and revascularization
Langabeer	2018	age, smoking, diabetes, prior CVD, prior stroke, heart failure, shock, length of stay, teaching, insurance, total ischemic time, door to balloon
Tang	2018	age, BMI, LVEF, serum creatinine, use of proton pump inhibitors, use of dual-antiplatelet therapy, previous PCI, diabetes mellitus, hypertension, previous stroke, current smoker, thrombocytopenia, use of femoral approach, use of intra-aortic balloon pump, and multivessel disease
Cenko	2019	age, family history of CAD, diabetes, hypertension, hypercholesterolemia, current smoking, former

		smoking, prior angina pectoris, prior myocardial infarction, prior PCI, prior CABG, peripheral artery disease, prior stroke, ST-segment elevation in anterior leads (at ECG), systolic blood pressure at baseline, heart rate at baseline, serum creatinine at baseline, Killip Class ⩾2
Hao	2019	Age, medical insurance status, acute heart failure, cardiogenic shock, cardiac arrest at admission, heart rate and systolic blood pressure, diabetes mellitus, smoking, history of CHD, heart failure, renal failure, and cerebrovascular disease, prehospital statin use, renal insufficiency, and transfer status.
Hannan	2019	age, STEMI location, heart rate, mean arterial pressure, history of hospitalization in last year, history of PCI, history of CABG surgery, septicemia/sepsis/systemic inflammatory response /shock, metastatic cancer/acute leukemia, diabetes with acute complications, end stage liver disease, inflammatory bowel disease, coagulation defects and other specified hematological disorders, dementia, polyneuropathy, muscular dystrophy, seizure disorders and convulsions, coma/brain compression/anoxic damage, cardiorespiratory failure and shock, congestive heart failure, specified heart arrhythmias, ischemic or unspecified stroke, hemiplegia/hemiparesis, vascular disease with complications, vascular disease without complications, aspiration and specified bacterial pneumonias, acute renal failure, chronic kidney disease, Stage 5, unspecified renal failure, nephritis, pressure ulcer of skin with partial thickness skin loss*, pressure pre-ulcer skin changes, chronic ulcer of skin except pressure ulcer, lower limb/amputation complications
Maznyczka	2019	NA
Stehli	2019	age, diabetes mellitus, eGFR, previous PCI and/or coronary artery bypass grafting, history of peripheral vascular disease and CVD, LVEF, out-of-hospital and in-hospital cardiac arrest, cardiogenic shock, and occurrence time of symptom onset
Burgess	2020	NA
Dharma	2020	NA
Kerkmanx	2020	NA
Siabani	2020	BMI⩾25, hypertension, diabetes, current smoking, hypercholesterolemia, congestive heart failure, Killip class (at first presentation) ⩾ II, symptom-to-balloon time> 360 min and door-to-balloon time > 90 min
Tai	2020	NA
Tizón	2020	age, diabetes mellitus, recruitment year, time from symptom onset to culprit coronary artery opening, and Killip class

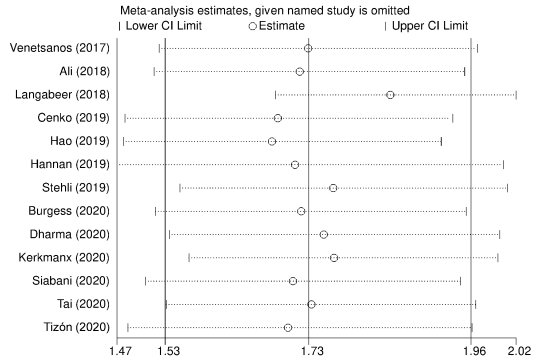
eTable 3 Assessment of study quality using Newcastle-Ottawa scale.

First Author	Year	Selection				Comparability	Outcome			Total points
		Representativeness of the exposed cohor	Selection of the no exposed cohort	Ascertainment of exposure to implants	Outcome of interest not present at start of study		Assessment of outcome	Follow-up long enough for outcomes to occur	Adequacy of follow-up	
Venetsanos	2017	*	*	*	*	**	*	\	*	8
Ali	2018	\	\	*	*	\	*	\	*	4
Langabeer	2018	*	*	*	*	*	*	\	*	7
Tang	2018	\	\	*	*	**	*	*	*	7
Cenko	2019	*	*	*	*	**	*	\	*	8
Hao	2019	*	*	*	*	**	*	\	*	8
Hannan	2019	*	*	*	*	**	*	\	*	8
Maznyczka	2019	\	\	*	*	\	*	*	*	5
Stehli	2019	*	*	*	*	**	*	\	*	8
Burgess	2020	\	\	*	*	**	*	*	*	7
Dharma	2020	\	\	*	*	*	*	*	*	6
Kerkmanx	2020	*	*	*	*	\	*	*	*	7
Siabani	2020	\	\	*	*	*	*	\	*	5
Tai	2020	\	\	*	*	**	*	*	*	7
Tizón	2020	*	*	*	*	**	*	*	*	9

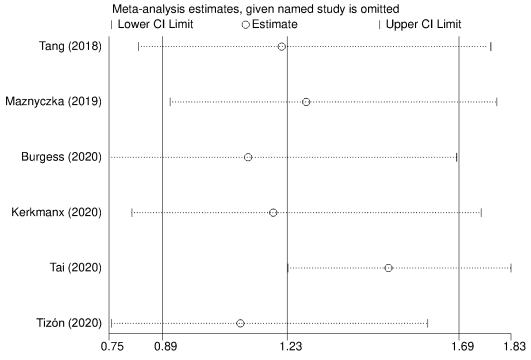
eFigure 1 Forest plots of relative risks of short-term all-cause mortality of studies with Newcastle-Ottawa scale >7 points and with ⩽7 points.



eFigure 2 Meta-influence analysis for unadjusted short-term mortality

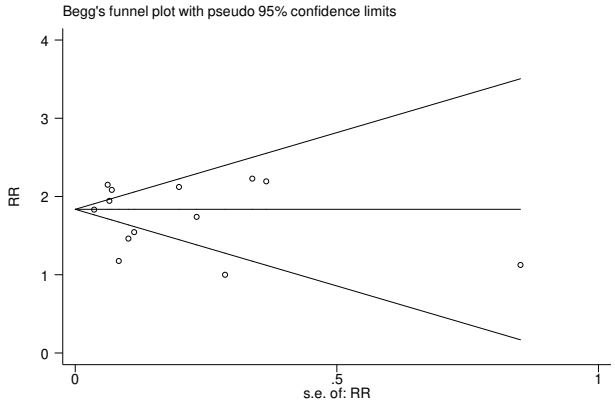
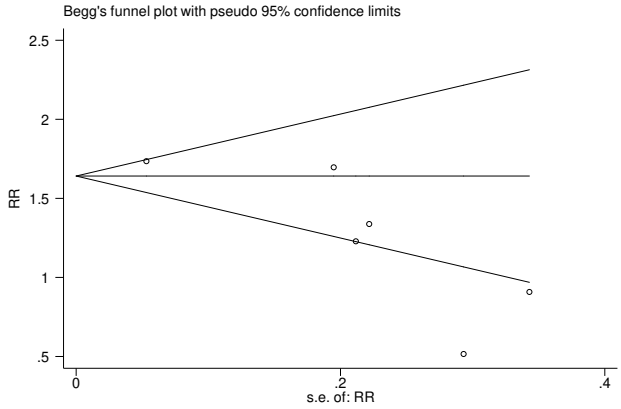


eFigure 3 Meta-influence analysis for unadjusted long-term mortality



eFigure 4 Funnel plots for publication bias for unadjusted short-term (A) and long-term (B) mortality

A



B